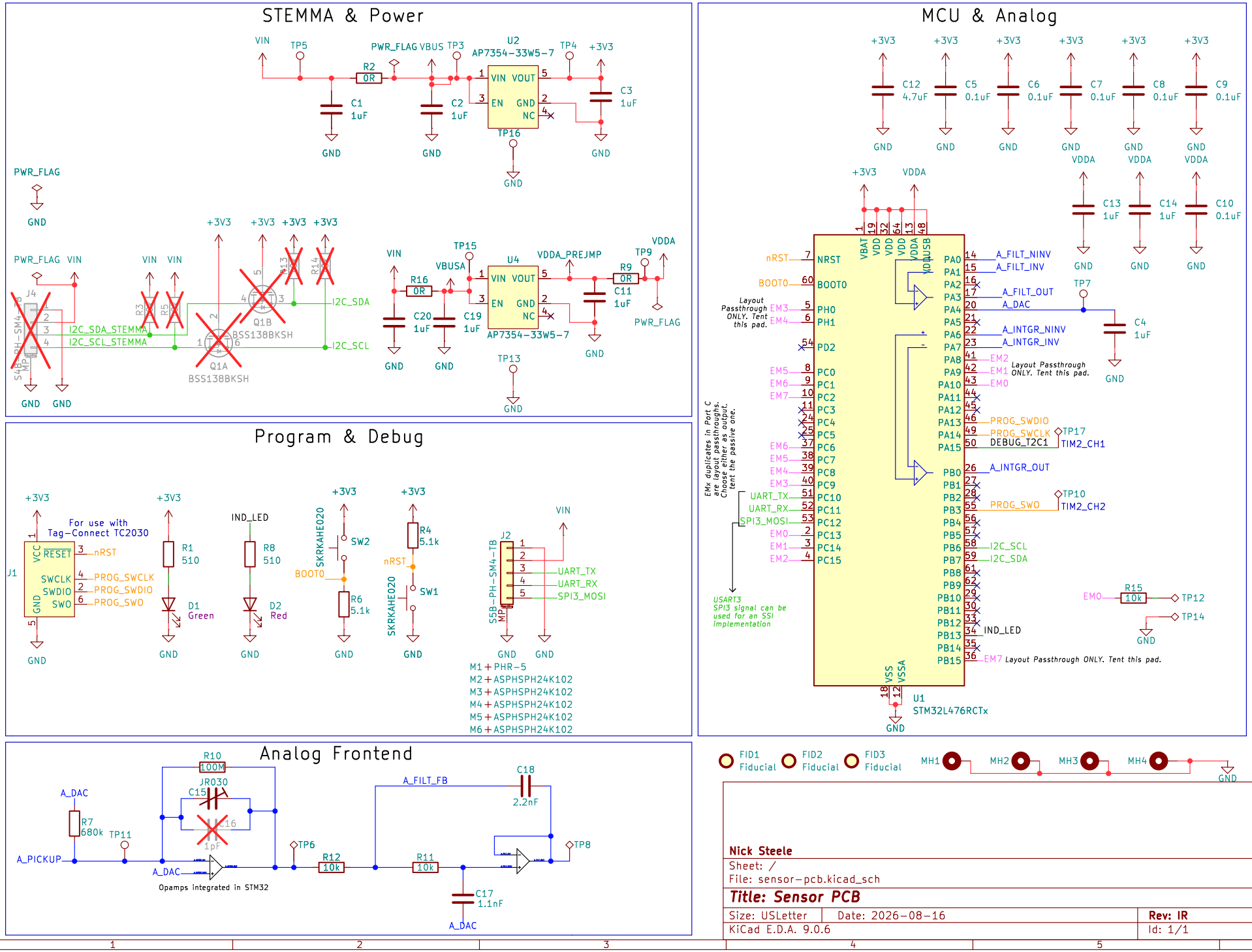

STEMMA & Power
VIN
TP5
PWR_FLAG
VBUS
TP3
U2
AP7354-33W5-7
TP4
+3V3
R2
0R
C1
1uF
C2
1uF
C3
1uF
TP16
GND
PWR_FLAG
PWR_FLAG VIN
VIN VIN
+3V3 +3V3 +3V3 +3V3
J4
SM4
R3
R5
R13
R14
Q1B
Q1A
BSS138BKSH
I2C_SDA_STEMMA
I2C_SCL_STEMMA
I2C_SDA
I2C_SCL
VIN
TP15
VBUSA
U4
VDDA_PREJMP
TP9
VDDA
R16
0R
R9
0R
C20
1uF
C19
1uF
C11
1uF
AP7354-33W5-7
PWR_FLAG
TP13
Program & Debug
+3V3
For use with
Tag-Connect TC2030
J1
VCC
RESET
SWCLK
SWDIO
SWO
GND
nRST
PROG_SWCLK
PROG_SWDIO
PROG_SWO
R1
510
D1
Green
IND_LED
R8
510
D2
Red
SKRKAHE020
SW2
BOOT0
R6
5.1k
R4
5.1k
nRST
SW1
J2
S5B-PH-SM4-TB
VIN
UART_TX
UART_RX
SPI3_MOSI
M1 + PHR-5
M2 + ASPHSPH24K102
M3 + ASPHSPH24K102
M4 + ASPHSPH24K102
M5 + ASPHSPH24K102
M6 + ASPHSPH24K102
Analog Frontend
R10
100M
JR030
C15
C16
1pF
A_DAC
R7
680k
TP11
A_PICKUP
A_DAC
Opamps integrated in STM32
TP6
R12
10k
A_FILT_FB
C18
2.2nF
R11
10k
TP8
C17
1.1nF
A_DAC
MCU & Analog
+3V3
C12
4.7uF
C5
0.1uF
C6
0.1uF
C7
0.1uF
C8
0.1uF
C9
0.1uF
GND
VDDA
C13
1uF
C14
1uF
C10
0.1uF
VBAT
VDD
VDDA
VDDUSB
nRST
BOOT0
Layout Passthrough ONLY. Tent this pad.
EM3
EM4
PH0
PH1
PD2
EM5
EM6
EM7
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
UART_TX
UART_RX
SPI3_MOSI
PC10
PC11
PC12
PC13
PC14
PC15
EMx duplicates in Port C are layout passthroughs. Choose either as output, tent the passive one.
USART3 SPI3 signal can be used for an SSI implementation
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
A_FILT_NINV
A_FILT_INV
A_FILT_OUT
A_DAC
A_INTGR_NINV
A_INTGR_INV
EM2
EM1
EM0
PROG_SWDIO
PROG_SWCLK
DEBUG_T2C1
TP17
TIM2_CH1
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
A_INTGR_OUT
TP10
PROG_SWO
TIM2_CH2
I2C_SCL
I2C_SDA
IND_LED
EM7
TP7
C4
1uF
EM0
R15
10k
TP12
TP14
VSS
VSSA
U1
STM32L476RCTx
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
MH1
MH2
MH3
MH4
Nick Steele
Sheet: /
File: sensor-pcb.kicad_sch
Title: Sensor PCB
Size: USLetter
Date: 2026-08-16
Rev: IR
KiCad E.D.A. 9.0.6
Id: 1/1